

Railway Reservation System

Python Programming Implementation

Project Description: This script implements a command-line interface for managing train bookings. It covers essential backend logic including seat inventory management, unique PNR generation, and cancellation handling.

Python Implementation

```
import random

class RailwayReservationSystem:
    def __init__(self):
        self.trains = {
            "101": {"name": "Express Express", "seats": 50, "fare": 500},
            "102": {"name": "Superfast Link", "seats": 30, "fare": 750},
            "103": {"name": "Coastal Voyager", "seats": 20, "fare": 1200}
        }
        self.bookings = {}

    def display_trains(self):
        print("\n--- Available Trains ---")
        for t_id, info in self.trains.items():
            print(f"ID: {t_id} | Name: {info['name']} | Seats: {info['seats']} | Fare: {info['fare']}")

    def book_ticket(self, train_id, passenger_name):
        if train_id in self.trains:
            if self.trains[train_id]['seats'] > 0:
                pnr = random.randint(10000, 99999)
                self.trains[train_id]['seats'] -= 1
                self.bookings[pnr] = {
                    "name": passenger_name,
                    "train": self.trains[train_id]['name'],
                    "fare": self.trains[train_id]['fare']
                }
            else:
                print("No seats available for this train.")
```

```

        print(f"\nSuccess! Ticket booked for {passenger_name}.")
        print(f"PNR: {pnr} | Train: {self.bookings[pnr]['train']} | Fare Paid: {fare}")
        return pnr
    else:
        print("\nSorry, no seats available in this train.")
else:
    print("\nInvalid Train ID.")

def cancel_ticket(self, pnr):
    if pnr in self.bookings:
        details = self.bookings.pop(pnr)
        # Find the train to restore seat
        for t_id, info in self.trains.items():
            if info['name'] == details['train']:
                info['seats'] += 1
                break
        print(f"\nBooking for PNR {pnr} ({details['name']}) has been cancelled.")
    else:
        print("\nInvalid PNR number.")

# Example Usage
if __name__ == "__main__":
    system = RailwayReservationSystem()
    system.display_trains()
    ticket_pnr = system.book_ticket("102", "Jubin")
    system.cancel_ticket(ticket_pnr)

```

Functionality Overview

- **Class-Based Structure:** Encapsulates data and methods within the `RailwayReservationSystem` class.
- **Data Structures:** Utilizes nested dictionaries to manage train availability and passenger bookings.
- **Error Handling:** Includes basic checks for invalid Train IDs and PNR numbers.